Unit 3

Hands on Exercises for Variance, Sampling, Covariance and Correlation

1. The delivery times of five online orders are:

20, 25, 30, 35, 40 minutes

Calculate the mean and population variance.

Given Data: **20, 25, 30, 35, 40**

Step 1: Calculate the mean

$\bar{X}$ = (20+25+30+35+40) / 5 = 150 / 5

Mean = 30

Step 2: Calculate deviations and squared deviations

X	X − Mean	(X − Mean)²
20	-10	100
25	-5	25
30	0	0
35	5	25
40	10	100
Total	**0**	**250**

Population variance:

$\sigma 2 = \sum(X-\mu)2 / N$

$\sigma 2 = 250 / 5$

Variance = 50

Answer:

- Mean = 30 minutes
- Population variance = 50 minutes²

2. A university has 5,000 students. A random sample of 100 students is selected to estimate the average daily study time. Explain why sampling is used instead of surveying all 5,000 students.

Given

Population = 5,000 students

Sample = 100 students

Surveying all 5,000 students would be a census, whereas selecting 100 students is sampling.

Sampling is preferred because:

- It saves time.
- It reduces cost and effort.
- Collecting data from 5,000 students may be difficult to manage.
- A properly selected random sample can provide a reliable estimate of the entire population.
- Data processing and analysis become easier.

Answer:
Sampling is used because studying 100 randomly selected students is much faster, cheaper, and easier than surveying all 5,000 students while still providing useful information about the population.

3. From a sample of 200 customers, 150 customers are satisfied with a product. Calculate the sample proportion of satisfied customers and express it as a percentage.

Total customers:

N = 200

Satisfied customers:

X = 150

Sample proportion:

$$\hat{p} = \frac{x}{n} = \frac{150}{200} = 0.75$$

Convert to percentage:

$0.75 \times 100 = 75\%$

Answer:

- Sample proportion = 0.75
- Percentage of satisfied customers = 75%

4. Calculate the covariance between the following variables:

X	Y
2	**5**
4	**7**
6	**9**
8	**11**

Determine whether the relationship is positive or negative.

Covariance

X	Y
2	5
4	7
6	9
8	11

First calculate the means:

$\bar{X} = (2+4+6+8) / 4 = 5$

$\bar{Y} = (5+7+9+11) / 4 = 8$

Now calculate:

X	Y	$X-\bar{X}$	$Y-\bar{Y}$	Product
2	5	-3	-3	9
4	7	-1	-1	1
6	9	1	1	1
8	11	3	3	9
				20

Using population covariance:

$Cov(X,Y) = \sum(X-\bar{X})(Y-\bar{Y}) / N = 20 / 4 = 5$

Interpretation:
Since covariance is positive, X and Y have a positive relationship. As X increases, Y also tends to increase.

If your class uses sample covariance, divide by $n-1$, giving $20/3=6.67$.

5. The following data represents hours of exercise (X) and calories burned (Y):

X	Y
1	**200**
2	**300**
3	**400**
4	**500**
5	**600**

Calculate the covariance and interpret the result.

Covariance Between Exercise and Calories

X (Hours)	Y (Calories)
1	200
2	300
3	400
4	500
5	600

Means:

$\bar{X} = (1+2+3+4+5) / 5 = 3$

$\bar{Y} = (200+300+400+500+600) / 5 = 400$

X	Y	$X-\bar{X}$	$Y-\bar{Y}$	Product
1	200	-2	-200	400
2	300	-1	-100	100
3	400	0	0	0
4	500	1	100	100
5	600	2	200	400
				1000

Population covariance:

$Cov(X,Y) = 1000 / 5 = 200$

Interpretation:
The covariance is +200, which indicates a positive relationship between exercise hours and calories burned. As exercise hours increase, calories burned also increase.

Sample covariance would be 1000/4=250.

6. Calculate Pearson's correlation coefficient for the following dataset and interpret the strength and direction of the relationship:

Study Hours (X)	Marks (Y)
1	45
2	50
3	55
4	65
5	70

Pearson's Correlation Coefficient

Study Hours (X)	Marks (Y)
1	45
2	50
3	55
4	65
5	70

We use:

$$r = \frac{n\sum XY - (\sum X)(\sum Y)}{\sqrt{[n\sum X^2 - (\sum X)^2][n\sum Y^2 - (\sum Y)^2]}}$$

Calculate the required values:

$$n = 5$$

$$\sum X = 15$$

$$\sum Y = 285$$

$$\sum X^2 = 55$$

$$\sum Y^2 = 16475$$

$$\sum XY = 925$$

Substitute:

$$r = \frac{5(925) - (15)(285)}{\sqrt{[5(55) - 15^2][5(16475) - 285^2]}}$$

$$r = \frac{4625 - 4275}{\sqrt{(275 - 225)(82375 - 81225)}}$$

$$r = \frac{350}{\sqrt{50 \times 1150}}$$

$$r = \frac{350}{239.79}$$

$$\boxed{r \approx 0.958}$$

Interpretation
The Pearson correlation coefficient is approximately +0.958.
Since r is close to +1, there is a very strong positive correlation between study hours and marks.

Conclusion:
Students who study more hours tend to obtain higher marks in this dataset.